Development Workflow and Current Progress: From Nonlocal Operator Audit to Clot Simulation

Establishing mechanically consistent nonlocal boundaries before quantitative thrombus failure prediction

1 COMPLETED

Theory & Discretization Audit

Nodal quadrature
Kernel / weights
Horizon handling

2 COMPLETED

BAQP Mechanics: Affine, Energy & First Variation

Affine kinematics
Energy + first variation
Spectral mechanics

3 COMPLETED

Independent Cross-Check: RKCMaterial / Peridynamics.jl

Local / author match
No clear porting error

Author force agreement:
5.9×10^{-15} relative

4 CURRENT

Finite-Domain Nonlocal Boundary Consistency

Affine ≠ equilibrium
Z / neighbor-Z channels
Cause: INCONCLUSIVE

5 NEXT

Fully Closed 3δ Force-Stencil Patch Test

Isolate central region
Close force dependencies

L075: 27 full families;
no closed force stencil

3δ: conservative bound

6 FUTURE

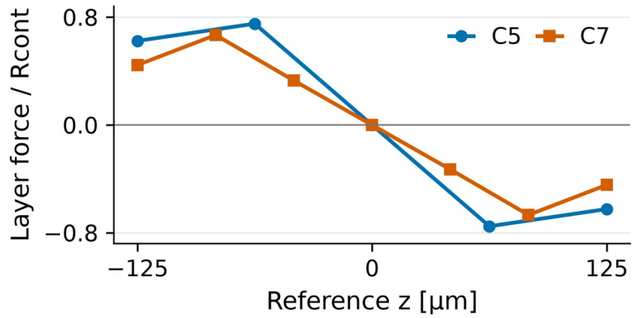
Physical Clot Simulation

Adhesion / detachment
Damage / rupture
Flow / microbubble load

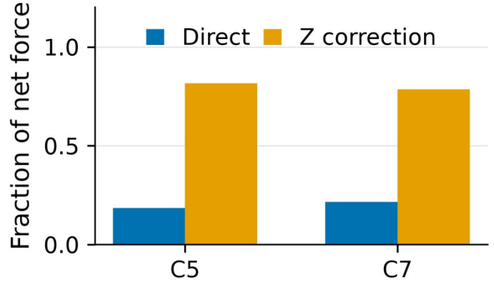
Future; not yet validated

BAQP + SPECTRAL MECHANICS + PETSc RSLS | Current audit: phase disabled

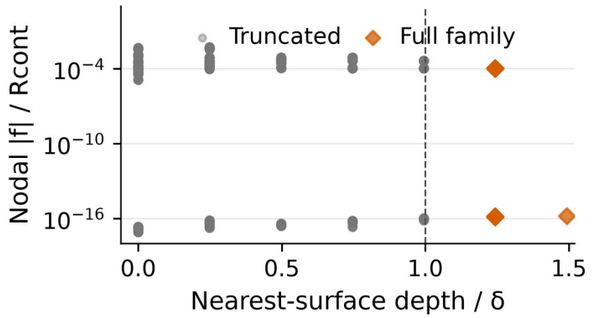
(a) Affine is not equilibrium



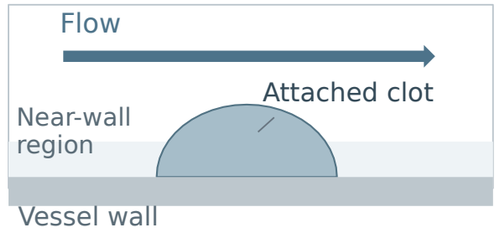
(b) Z-mediated force transfer



(c) L = 0.75 mm: full family



TARGET APPLICATION



Target application:
mechanically interpretable
clot-wall failure

Activation ≠ fracture • Stationary ≠ stable • Self-family complete ≠ force-stencil isolated

Verified Mechanics › **Boundary Root-Cause Resolution** › **Physical Clot Modeling**

Current bottleneck: separating finite-domain boundary influence from true bulk operator behavior.

Next: fully closed 3δ force-stencil affine patch test.